System Administration and Maintenance

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IP Address

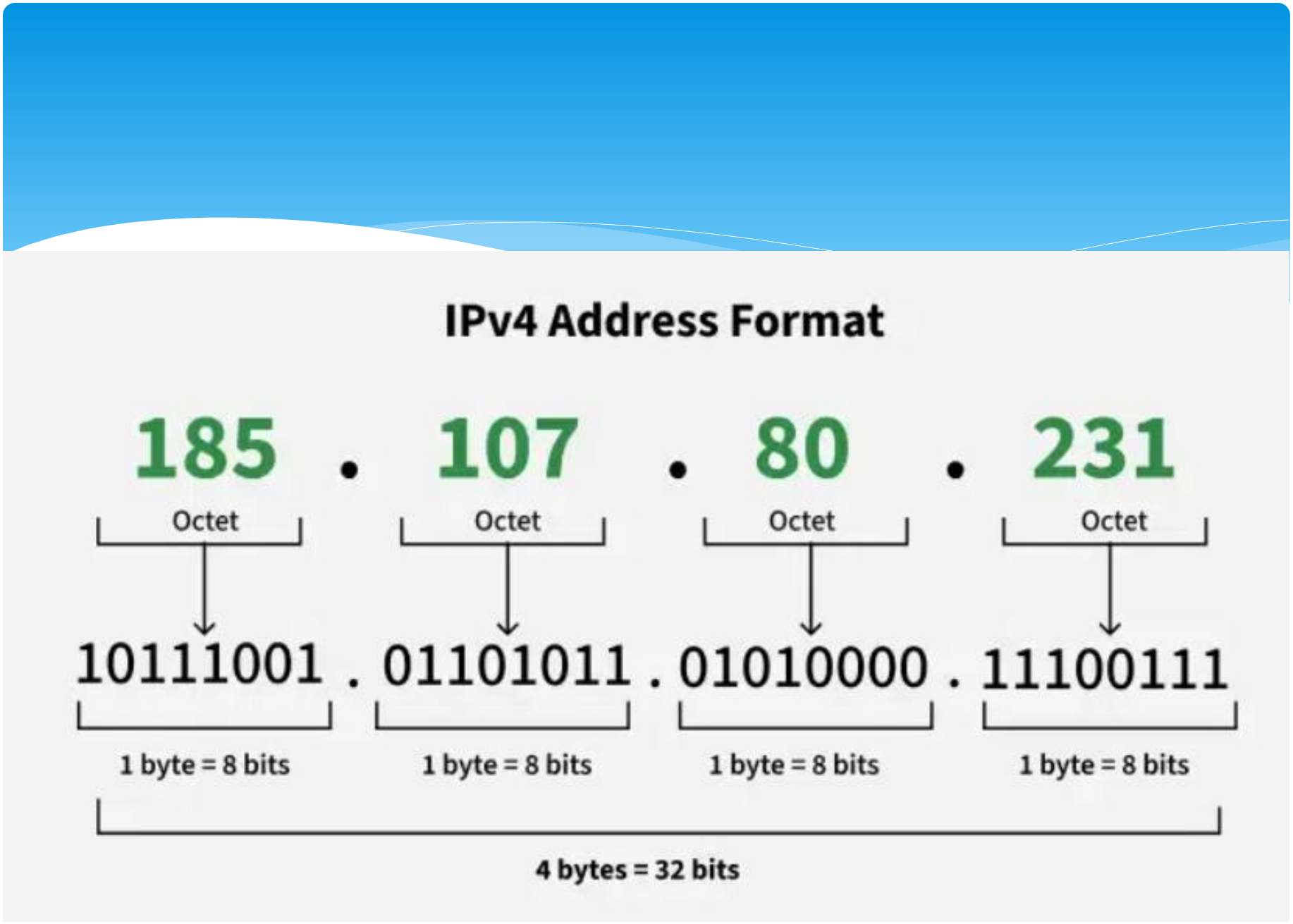
- An IP address or **Internet Protocol** address.
- It is a **unique string** of numbers assigned to each device connected to a computer network.
- Uses the Internet Protocol for communication.
- It serves as an identifier that allows devices to send and receive data over the network.
- Ensuring that this data reaches the correct destination.

IP Versions **(IPv4 vs IPv6)**

IPv4

- ✓ This is the most common form of IP Address.
- ✓ It consists of four sets of numbers separated by dots.
- ✓ Eg:

  192.158.1.38 Each set of numbers can range from **0 to 255.**
- ✓ This format can support over **4 billion unique addresses**.
- ✓ **Four Octets**:
  - Each octet represents eight bits, or a byte, and can take a value from **0 to 255**.
  - This range is derived from the possible combinations of eight bits (**2^8 = 256 combinations**).
- ✓ Example of IPv4 Address: **192.168.1.1**
  - **192** is the first octet
  - **168** is the second octet
  - **1** is the third octet
  - **1** is the fourth octet


IPv4 Address Format
185 . 107 . 80 . 231
Octet
Octet
Octet
Octet
10111001 . 01101011 . 01010000 . 11100111
1 byte = 8 bits
1 byte = 8 bits
1 byte = 8 bits
1 byte = 8 bits
4 bytes = 32 bits

IPv6

- IPv6 addresses were created to deal with the shortage of IPv4 addresses.
- Use **128 bits instead of 32**, offering a vastly greater number of possible addresses.
- These addresses are expressed as eight groups of **four hexadecimal digits**.
- Each group representing **16 bits**. The groups are separated by colons.
- **Example of IPv6 Address** :

  2001:0db8:85a3:0000:0000:8a2e:0370:7334
- Each group (like **2001**, **0db8**, **85a3**, etc.) represents a 16-bit block of the address.

Public IP vs Private IP

Public IP Addresses

- A Public IP address is assigned to every device that directly accesses the internet.
- This address is unique across the entire internet.
- Characteristics and uses of public IP addresses:
  - **Uniqueness:**
    - Each public IP address is globally unique. No two devices on the internet can have the same public IP address at the same time.
  - **Accessibility:**
    - Devices with a public IP address can be accessed directly from anywhere on the internet, assuming no firewall or security settings block the access.
  - **Assigned by ISPs:**
    - Public IP addresses are assigned by Internet Service Providers (ISPs).
    - When you connect to the internet through an ISP, your device or router receives a public IP address.
  - **Types:**
    - Public IP addresses can be static (permanently assigned to a device) or dynamic (temporarily assigned and can change over time).

Private IP Addresses

- Private IP addresses are used within private networks like home networks, office networks, etc
- Those are not routable on the internet.
- This means that devices with private IP addresses cannot directly communicate with devices on the internet without a translating mechanism like a router performing **Network Address Translation (NAT).**
- Characteristics and uses of Privet IP addresses:
  - **Not globally unique:**
    - Private IP addresses are only required to be unique within their own network.
    - Different private networks can use the same range of IP addresses without conflict.
  - **Local communication:**
    - These addresses are used for communication between devices within the same network.
    - They cannot be used to communicate directly with devices on the internet.
  - **Defined ranges:**
    - The **Internet Assigned Numbers Authority (IANA)** has reserved specific IP address ranges for private use:
      - ✓ **IPv4:** 10.0.0.0 to 10.255.255.255, 172.16.0.0 to 172.31.255.255, 192.168.0.0 to 192.168.255.255
      - ✓ **IPv6:** Addresses starting with FD or FC

THANK YOU